

A Signal-Agnostic Ingestion and Radiomic Feature-Extraction Pipeline for a Multi-Family Clinical Decision-Support System

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ABSTRACT

Clinical decision-support systems (CDSS) are usually built around a single data modality, which forces institutions to maintain a separate software stack for every family of biomedical signal. We describe *BioStatusIA*, an open, modular pipeline that ingests heterogeneous biomedical data through a single entry point and adapts its processing to whatever is technically feasible. The system automatically classifies an arbitrary file or folder into one of nine operating modes and routes it to one of three physical signal families—one-dimensional physiological time series (F1: ECG, EEG, EMG, EOG, PPG, spirometry), two-dimensional DICOM images (F3: radiography, mammography, ultrasound), and three-dimensional volumes (F4: CT, MR, PET/SPECT)—plus a tabular branch for pre-computed clinical features. A single normalized data contract, *SinalNormalizado*, decouples the readers from the domain-specific feature extractors, so that morphological, textural (GLCM), spectral (Welch PSD) and radiomic biomarkers are computed through a uniform interface regardless of the input. We report the behaviour of the pipeline on a real breast-ultrasound dataset (780 images), a clinical tabular dataset, and synthetic exemplars of each family, and we critically discuss the engineering trade-offs of a signal-agnostic architecture. The pipeline is released as pure, side-effect-free functions to encourage reproducibility and reuse.

Index Terms— clinical decision support, biomedical signal processing, radiomics, feature extraction, DICOM, medical imaging, software architecture, reproducibility.

1. INTRODUCTION

Modern clinical practice generates an unusually broad spectrum of digital signals: an electrocardiogram is a one-dimensional voltage series, a chest radiograph is a two-dimensional pixel array stored in DICOM, a computed-tomography study is a three-dimensional voxel volume, and a laboratory panel is a row of scalar features. Each of these modalities has, historically, its own analysis software, its own file formats, and its own community of tools [1–3]. For a small research group—such as a graduate laboratory building a proof-of-concept CDSS—this fragmentation is a serious obstacle: every new modality means a new ingestion path, a new set of dependencies, and a new opportunity for silent failure.

The central engineering hypothesis of this work is that a large fraction of the early processing stages—file typing, reading, normalization, statistical characterization, and biomarker extraction—can be expressed *once* behind a common data contract, and that the parts that genuinely differ between modalities can be isolated in thin, replace-

able extractors. We call the resulting system *BioStatusIA*. Its design principle is deliberately permissive: *the system accepts any input in scope, always runs a statistical analysis on whatever it receives, and adapts the remainder of the pipeline to what is feasible*. Supervised model training is attempted only when labels and a sufficient number of examples are present; otherwise the system still returns a full descriptive characterization.

This paper concentrates on the *ingestion and feature-extraction* subsystem—the part of *BioStatusIA* that turns a raw, possibly unlabelled, possibly unknown file into a structured vector of biomarkers. The classification, model-selection and reporting layers are the subject of a companion paper. Our contributions are: (i) an automatic, filesystem-level modality detector with nine operating modes; (ii) a single normalized data contract that decouples readers from extractors across three physical signal families and a tabular branch; (iii) a set of domain-specific but uniformly exposed biomarker extractors; and (iv) an honest empirical and architectural evaluation, including the limitations that a signal-agnostic design imposes.

2. RELATED WORK

Radiomics—the high-throughput extraction of quantitative descriptors from medical images—has become a standard bridge between raw pixels and machine learning [4, 5]. Mature libraries exist for individual modalities: MNE-Python and WFDB for physiological time series [1], pydicom for DICOM [2], and NiBabel and SimpleITK for volumetric formats [3]. General-purpose scientific stacks such as scikit-image and OpenCV provide the texture and morphology primitives [6, 7]. What is comparatively rare in the literature is an integration layer that treats these modalities as interchangeable inputs to a *single* decision-support front end. Commercial PACS and vendor CDSS products are typically modality-locked, and open academic systems tend to target one signal family at a time. *BioStatusIA* occupies this gap: rather than proposing a new descriptor, it proposes an *architecture* in which existing descriptors are exposed through one contract and one automatic dispatcher.

3. SYSTEM ARCHITECTURE

3.1. Design constraints

The pipeline is written as a collection of *pure functions*: each stage receives data and returns data, with no hidden input/output. All persistence (database writes, HTML rendering, file handling) is concentrated in the web layer, so the scientific core can be unit-tested and reused in isolation. This separation is the main reason the same extraction code

Table 1: The nine in-scope operating modes and the family they route to.

Mode	Trigger	Family
imagem_unica	one raster image	Image
imagens_soltas	unlabelled image folder	Image
dataset_rotulado	benign/+malignant/	Image
tabular	CSV/TSV/TXT	Tabular
multimodal	images + tabular	Image+Tab
senal_temporal	.dat/.edf/.mat/...	F1
imagem_dicom_2d	single .dcm	F3
volume_3d	.nii/.mha, ≥ 10 .dcm	F4
multimodal_expandido	mixed families	F1+F3+HeartRate

can be driven both by an interactive web server and by a batch command-line entry point.

3.2. Automatic modality detection

The dispatcher, `detectar_estrutura`, receives a path—which may be a single file, a folder, or an extracted archive—and returns one of nine in-scope operating modes. For a single file it tests, in order, the predicates for common images, tabular files, physiological time series, single-slice DICOM, and 3-D volumes. For a directory it recursively inventories the file types present and resolves multimodal combinations; folders whose sub-directories carry recognizable class names (e.g. `benign/`, `malignant/`) are promoted to a *labelled dataset* mode. A folder of DICOM files is disambiguated by cardinality: ten or more slices are read as a 3-D series, fewer as a set of 2-D images. Inputs outside the supported scope resolve to an explicit *invalid* mode rather than failing silently. Table 1 summarizes the modes and the physical family each triggers.

3.3. The normalized data contract

The keystone of the design is a single dataclass, `SinalNormalizado`, returned by every reader:

```
familia,      tipo,      dados (ndarray),
taxa_amostragem, canais, metadados,
caminho_original, dados_viz (a downsampled
trace of  $\leq 2000$  points for the browser).
```

All reading passes through a universal dispatcher, `carregar_sinal`, which selects the correct reader by extension—WFDB for `.dat/.hea`, MNE for `.edf/.bdf`, SciPy for `.mat`, an XML parser for spirometry exports, `pydicom` for `.dcm`, and NiBabel/SimpleITK for volumes—and returns the same structure in every case. Because the extractors consume only `SinalNormalizado`, adding a new file format requires writing one reader and touching nothing downstream. This contract is therefore the explicit boundary between the “read” and “understand” halves of the system, and we treat it as inviolable.

4. FEATURE EXTRACTION

Each physical family has a dedicated extractor that consumes the normalized signal and emits a dictionary of biomarkers grouped by domain.

4.1. F1 — physiological time series

Time-domain descriptors (RMS, mean, standard deviation, skewness, kurtosis, peak-to-peak amplitude, SNR in dB) are complemented by frequency-domain descriptors obtained from a Welch power-spectral-density estimate: dominant frequency, spectral centroid, and normalized band powers. Signal-type-specific analyses are dispatched from the declared type: for ECG the extractor band-pass-filters the highest-variance channel (0.5–40 Hz, 4th-order zero-phase Butterworth), detects R-peaks with a physiologically motivated minimum inter-beat distance, and derives heart rate and the classical heart-rate-variability indices RMSSD, SDNN and pNN50; for EEG it computes relative delta/theta/alpha/beta/gamma band powers and the alpha/beta ratio; for EMG, envelope RMS and median frequency; for spirometry, FVC, FEV1, the FEV1/FVC ratio and PEF.

4.2. F3 — 2-D DICOM

On single-slice DICOM the extractor computes nine imaging biomarkers—two morphological (circularity, solidity from the largest external contour), four textural from the gray-level co-occurrence matrix (contrast, homogeneity, energy and a histogram entropy), and three intensity descriptors (SNR, skewness, kurtosis)—augmented by DICOM-specific quantities such as the fraction of high-density pixels, mean gradient, modality tag and pixel spacing, with automatic Hounsfield windowing driven by the header’s window centre/width.

4.3. F4 — 3-D volumes

Volumes are re-oriented to an axial-first convention and characterized by global intensity statistics (mean, dispersion, median, skewness, kurtosis, 5th/95th percentiles, count of high-intensity voxels), GLCM texture computed independently on the three orthogonal planes, and 3-D morphology (lesion volume in mm^3 , sphericity, bounding-box axes). A helper renders an arbitrary axial slice to base64 for visualization without exposing a separate image route.

4.4. Common image radiomics and the tabular branch

For ordinary rasters the same nine morphological/textural/intensity biomarkers as F3 are computed after an adaptive pre-processing stage that chooses denoising, normalization and equalization from a lightweight analysis of the base (intensity, contrast, outliers, normality and size consistency). The tabular branch parses arbitrary CSV/TSV/TXT files: it sniffs the delimiter, tolerates UTF-8 and Latin-1 encodings, and detects the label column either by name (`label`, `class`, `diagnosis`, `target`, ...) or, failing that, as the last column with between two and ten distinct values; binary labels are mapped to $\{0, 1\}$ through a keyword table.

5. EXPERIMENTAL SETUP AND RESULTS

We exercised the pipeline on three kinds of input to characterize its behaviour rather than to claim state-of-the-art accuracy. All experiments use Python 3.11, scikit-learn,

scikit-image, OpenCV, SciPy, MNE, pydicom and NiBabel; random seeds are fixed for reproducibility.

5.1. Real breast-ultrasound imaging (BUSI)

On the Breast Ultrasound Images dataset [8], the pipeline autonomously analyzed the base of 780 images (estimated mean intensity 83.6, mean contrast 51.5, Shapiro $p=0.021$, inconsistent sizes) and selected a Gaussian-denoising / min-max-normalization / 256×256 strategy. Biomarker extraction succeeded on all 780 images (570 benign, 210 malignant). The extraction stage is the load-bearing part evaluated here; the downstream six-model comparison (reported in full in the companion paper) yields, for the best model, an accuracy of 0.814 and an ROC-AUC of 0.800, with a recall of 0.452 that transparently reflects the untreated class imbalance of this branch. We regard these realistic—not perfect—numbers as the honest operating point of the extraction pipeline on genuine clinical data.

5.2. Clinical tabular data

On a 50-row breast-cancer panel with 30 numerical features, the tabular parser correctly identified the diagnosis label, mapped B/M to 0/1, produced a balanced 25/25 split and computed univariate statistics for all features (e.g. mean radius 15.32 ± 3.90 , mean area 770.6 ± 389.8). The very small sample size makes any downstream accuracy figure optimistic, a point we emphasize rather than hide.

5.3. Synthetic exemplars of each family

To smoke-test the F1/F3/F4 readers and extractors end-to-end we generated small synthetic sets (14 exemplars each). The dispatcher correctly identified a MATLAB ECG signal and produced time and frequency feature groups; a synthetic CT DICOM produced morphological, GLCM, intensity and DICOM-radiomic groups (15-dimensional vector); and a synthetic NIFTI volume produced global, 3-D texture and 3-D morphology groups (13-dimensional vector). These runs confirm the plumbing of every family but, being synthetic and tiny, are explicitly *not* evidence of diagnostic performance.

6. DISCUSSION

The results support the architectural claim: a single contract and a single dispatcher are sufficient to carry four data families from raw file to structured biomarkers, and the pure-function decomposition makes each stage independently testable. The design also exposes clear limitations. First, a signal-agnostic front end trades depth for breadth: the GLCM descriptor is computed at a single distance and orientation and is therefore not rotation-invariant, and the physiological analyses use robust but basic estimators rather than modality-optimized toolchains. Second, the extractors emit vectors of heterogeneous dimensionality (nine to fifteen features across families), so cross-family model reuse requires an explicit alignment step. Third, the permissive philosophy that lets the system “always run something” can mask degenerate inputs; the single-image

smoke test, for instance, produces biomarkers that are individually valid but statistically meaningless. We view these not as defects to conceal but as the natural cost of generality, and as a roadmap: multi-orientation Haralick features, a per-family feature schema, and input-quality gating are concrete next steps.

7. ETHICAL CONSIDERATIONS

BioStatusIA is a decision-*support* tool. Every visual output and every generated report carries a mandatory notice that its results do not replace assessment by a licensed physician. The extraction pipeline computes descriptive biomarkers only; it makes no autonomous clinical decision, and no result should be acted upon without professional review. Datasets used here are public and de-identified.

8. CONCLUSION

We presented the ingestion and feature-extraction core of BioStatusIA, a signal-agnostic CDSS pipeline that detects its input automatically, normalizes four data families behind one contract, and extracts domain-appropriate biomarkers through a uniform interface. Empirical runs on real imaging, real tabular data and synthetic signals demonstrate that the architecture is sound and reproducible, while an honest discussion of its limitations frames the path toward richer descriptors and stronger input validation. The companion paper builds the clinically-enriched AutoML and multi-agent reporting layers on top of this foundation.

REFERENCES

- [1] A. Gramfort *et al.*, “MEG and EEG data analysis with MNE-Python,” *Frontiers in Neuroscience*, vol. 7, art. 267, 2013.
- [2] D. Mason, “SU-E-T-33: Pydicom: an open source DICOM library,” *Medical Physics*, vol. 38, no. 6, p. 3493, 2011.
- [3] M. Brett *et al.*, “nibabel: Access a multitude of neuroimaging data formats,” Zenodo, 2020.
- [4] R. J. Gillies, P. E. Kinahan, and H. Hricak, “Radiomics: images are more than pictures, they are data,” *Radiology*, vol. 278, no. 2, pp. 563–577, 2016.
- [5] R. M. Haralick, K. Shanmugam, and I. Dinstein, “Textural features for image classification,” *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-3, no. 6, pp. 610–621, 1973.
- [6] S. van der Walt *et al.*, “scikit-image: image processing in Python,” *PeerJ*, vol. 2, p. e453, 2014.
- [7] F. Pedregosa *et al.*, “Scikit-learn: Machine learning in Python,” *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.
- [8] W. Al-Dhabyani, M. Goma, H. Khaled, and A. Fahmy, “Dataset of breast ultrasound images,” *Data in Brief*, vol. 28, p. 104863, 2020.